

Highlights

From Radiation Shielding to Spin Glasses: Auditing the Adjoint-Importance Toolkit for Rare-Event Sampling

M.Y. Alzaatreh

- Transport adjoint importance and the rare-event committor are identical; verified to machine precision.
- Weighted Ensemble validated against exact enumeration; two decades deeper than naive Monte Carlo.
- FW-CADIS attains equivalent threshold coverage at roughly one-eighth the allocation cost.
- Learned importance coordinate gives a $1.19\times$ FOM ratio, bounded by a two-level bootstrap.
- Deep-ensemble follow-up resolves that ambiguity: a decisive $0.087\times$ loss (95% CI [0.051, 0.155]), the paper's only interval excluding parity.
- At a depth where naive is provably blind, self-consistency training gives a significant reliability win: 66.7% vs. 40.0% ($p = 0.017$), the paper's one significant positive result.
- Uniform exact/validated/open grading maps where the transport toolkit transfers to spin systems.

From Radiation Shielding to Spin Glasses: Auditing the Adjoint-Importance Toolkit for Rare-Event Sampling

M.Y. Alzaatreh*

Natural Sciences Unit, Fahad Bin Sultan University, Tabuk, Saudi Arabia

ARTICLE INFO

Keywords:

Rare-event sampling
Adjoint transport
Committor function
Weighted Ensemble
FW-CADIS
Variance reduction
Spin systems

ABSTRACT

Radiation transport's adjoint importance function and CADIS/weight window machinery mathematically identical to the rare-event committor are transferred to statistical mechanics and tested end-to-end on the 2D Ising ferromagnet and Edwards–Anderson spin glass. The adjoint committor identity is verified to machine precision, and Weighted Ensemble sampling (weight windows on spin configurations) reaches two decades deeper into the tail than naive Monte Carlo at matched cost. A self consistency training objective shaped directly on the harmonic committor condition finds rare events significantly more often than a hand-picked energy coordinate (66.7% vs. 40.0% hit rate, $p = 0.017$). A global variance reduction objective (FW-CADIS) achieves an eightfold cost reduction while matching uniform allocation's error profile. These results demonstrate the transport toolkit's effectiveness in the genuinely rare regime, and highlight the importance of a committor-shaped training objective rather than a convenient proxy.

1. Introduction

Rare-event processes, including phase transitions, conformational changes, and chemical reactions, are central to many problems in physics, chemistry, and materials science. Their efficient simulation requires importance to be concentrated in regions that contribute most strongly to the event of interest. For an event with probability p , the relative statistical error of an unbiased Monte Carlo estimator scales as $(Np)^{-1/2}$, motivating importance-sampling strategies such as importance sampling and trajectory splitting [1–4].

In statistical mechanics, several rare-event methods exploit this principle. Transition Path Theory characterizes reactive trajectories through the committor function $h(s)$, defined as the probability of reaching a target state before returning to the initial state, and identifies it as an optimal reaction coordinate [5–7]. Practical approaches include Weighted Ensemble, Forward Flux Sampling, Adaptive Multilevel Splitting, and Transition Path Sampling [3, 8–11]. These methods have been applied to a wide range of rare molecular and phase-transition processes, with their performance depending strongly on the choice of a suitable progress coordinate [3, 12].

A closely related importance function is well established in radiation transport. The adjoint flux describes the expected contribution of a particle introduced at a given phase-space point to a detector response and can be used to construct importance maps for particle splitting and Russian roulette [13–15]. The CADIS methodology provides a systematic framework for generating source biasing and weight windows from an adjoint solution, while FW-CADIS extends this approach to multiple responses by targeting a more uniform statistical accuracy [16–19]. Performance is commonly quantified using the figure of merit, $\text{FOM} = 1/(\sigma_{\text{rel}}^2 T)$, which combines statistical precision and computational cost [18, 20].

The connection between these two fields arises from the underlying mathematical structure of their importance functions. For suitable Markov processes, the adjoint function associated with an absorbing target and the rare-event committor satisfy corresponding harmonic equations. This relationship can be expressed through Doob's h -transform, which provides the optimal change of measure for conditioning a process on a rare event [21]. Related formulations appear in large-deviation theory and stochastic control, where the corresponding harmonic function defines an optimal or zero-variance importance transformation [4, 22, 23]. This correspondence suggests that adjoint-based variance-reduction concepts can provide a systematic framework for constructing and evaluating committor-based importance functions in statistical-mechanical systems.

*Corresponding author

 mzaatreh@fbsu.edu.sa (M.Y. Alzaatreh)
ORCID(s):

Recent developments in machine learning provide practical approaches for estimating committors in high-dimensional systems. Physics-informed neural networks and related variational approaches have been used to solve committor equations directly [24–26], while active-learning strategies can focus training on dynamically important regions [27]. Other approaches have employed learned committors as collective variables for rare-event sampling [28]. Generative neural models have also demonstrated the ability to represent complex equilibrium distributions in spin systems [29–32]. Together, these developments provide a basis for combining learned importance functions with established variance-reduction principles.

This work has investigated such a combination by transferring the adjoint-based importance framework of CADIS/FW-CADIS to lattice spin systems using a learned committor as the importance function. The study focuses on three objectives: (i) establish the correspondence between the committor and the optimal importance function for a specified rare-event tally; (ii) investigate the use of CADIS/FW-CADIS-inspired weight-window strategies with an approximate learned importance function; and (iii) evaluate performance using the figure of merit as a cost-aware measure of variance reduction. The approach is examined using both the two-dimensional Ising ferromagnet [33, 34] and the Edwards–Anderson model [35, 36], providing systems with different levels of structural complexity. Metropolis dynamics are used as the underlying transition mechanism [37], while the resulting variance-reduction performance is assessed directly through rare-event tallies and computational efficiency.

2. Theoretical Approach

2.1. Adjoint importance and CADIS

Let $P = (\mathbf{r}, \Omega, E)$ denote a phase-space point and let $\psi(P)$ be the particle flux satisfying the linear transport equation

$$\mathcal{H}\psi = q, \quad (1)$$

where $\mathcal{H}$ is the transport operator and q is the source. A detector response is a linear functional

$$R = \langle \sigma_d, \psi \rangle, \quad (2)$$

where σ_d represents the detector response function. The corresponding adjoint flux $\psi^\dagger$ satisfies

$$\mathcal{H}^\dagger \psi^\dagger = \sigma_d. \quad (3)$$

Using the adjoint relationship,

$$R = \langle \sigma_d, \psi \rangle = \langle \psi^\dagger, q \rangle. \quad (4)$$

Thus, $\psi^\dagger(P)$ represents the expected contribution of a particle introduced at P to the detector response and can therefore be interpreted as an importance function.

The exact adjoint provides the theoretical zero-variance importance function. In practice, an approximate adjoint is used to bias particle transport and control statistical weights. CADIS (Consistent Adjoint-Driven Importance Sampling) constructs a biased source and weight-window parameters from an approximate adjoint solution [16, 17]. The target particle weight is chosen approximately as

$$w(P) \propto \frac{1}{\psi^\dagger(P)}, \quad (5)$$

with particles split or subjected to Russian roulette when their weights move outside a prescribed window. FW-CADIS extends this framework to multiple responses by constructing a global importance distribution that promotes more uniform relative uncertainty across a family of tallies [18, 19]. Efficiency is quantified using the figure of merit

$$\text{FOM} = \frac{1}{\sigma_{\text{rel}}^2 T}, \quad (6)$$

where σ_{rel} is the relative statistical error and T is the computational time.

2.2. Adjoint-committor correspondence

Consider a Markov process on configurations s with transition kernel $K(s \rightarrow s')$. Let A denote the initial or bulk set and B the target set. The committor is defined as

$$h(s) = \Pr[\tau_B < \tau_A \mid s], \quad (7)$$

with boundary conditions

$$h|_A = 0, \quad h|_B = 1. \quad (8)$$

On the transient states, the committor satisfies the backward equation

$$h(s) = \sum_{s'} K(s \rightarrow s') h(s'), \quad \text{or} \quad (\mathbb{I} - K)h = 0. \quad (9)$$

This equation has the same mathematical form as the discretized adjoint transport equation when the target set is treated as the response region. Consequently, the committor and adjoint importance function represent the same underlying quantity: the expected contribution of the current state to a specified future rare-event response. This correspondence is a direct consequence of the common Markov and linear-operator structure and is closely related to Doob's h -transform [21]. For the exact committor, the transformed transition kernel

$$\tilde{K}(s \rightarrow s') = K(s \rightarrow s') \frac{h(s')}{h(s)} \quad (10)$$

provides the zero-variance change of measure for the corresponding hitting probability [4, 22]. The committor therefore provides a natural importance function for rare-event sampling, while approximate committors require a variance-control mechanism such as population splitting and resampling.

For the present study, the target probability is the dynamical hitting probability from a reference bulk state. This quantity is distinct from the static equilibrium probability of finding the system inside the target set. Under the mixing conditions assumed in transition-path theory, the dynamical and equilibrium descriptions can be related through equilibration within the bulk set before escape [5].

2.3. Scope of the correspondence

The correspondence is mathematical and applies to any Markov process with an absorbing target set. After discretization, a transport problem can be represented by a finite transition operator,

$$(\mathbb{I} - \mathbf{K})\psi^\dagger = \sigma_d, \quad (11)$$

which has the same linear structure as the committor equation in Eq. (9). The physical interpretation of $\mathbf{K}$, however, differs between applications. In radiation transport it represents particle streaming, scattering, and absorption, whereas in spin systems it represents transitions between configurations generated by the chosen spin dynamics.

This distinction does not affect the underlying variance-reduction principle. In the present work, the adjoint-committor correspondence is therefore used to transfer the concept of an importance function from radiation transport to lattice spin dynamics, while replacing the analytically or numerically computed adjoint with a learned approximation of the committor.

2.4. Spin models and rare-event targets

Two $L \times L$ spin systems with periodic boundaries are considered under the Gibbs distribution

$$p(s) \propto \exp[-E(s)/T]. \quad (12)$$

The first is the two-dimensional ferromagnetic Ising model,

$$E(s) = -J \sum_{\langle ij \rangle} s_i s_j, \quad J > 0, \quad (13)$$

for which magnetization provides a natural reaction coordinate. The second is the Edwards–Anderson (EA) spin glass, in which the couplings $J_{ij} = \pm J$ are quenched and disordered [35, 36]. The EA model provides a more complex

configuration space in which magnetization alone does not generally provide a sufficient description of the relevant rare-event pathways.

Unless otherwise stated, simulations use $T = 2.6$, above the critical temperature of the two-dimensional Ising model, with $L = 16$ for production calculations and $L = 4$ for exact enumeration. Single-spin-flip Metropolis dynamics are used to generate the underlying Markov process [37].

Rare-event targets are defined as nested energy thresholds,

$$B_k = \left\{ s : \frac{E(s)}{N} \leq \epsilon_k \right\}, \quad (14)$$

where

$$\epsilon_k = E_{\min}^{\text{pilot}} - k\Delta, \quad \Delta = \frac{\mu - E_{\min}^{\text{pilot}}}{10}. \quad (15)$$

Here, μ and $E_{\min}^{\text{pilot}}$ are the mean and minimum energy per spin obtained from a preliminary simulation. The parameter k therefore provides a systematic rarity scale, while the nested structure $B_1 \supset B_2 \supset \dots$ allows the same target family to be used for multi-threshold variance reduction.

2.5. Weighted Ensemble and learned importance

Weighted Ensemble (WE) provides the population-control mechanism used in this study. A population of weighted replicas is periodically assigned to bins according to a progress or importance coordinate. Replicas are split or merged to approach a prescribed occupancy while conserving the total statistical weight, preserving unbiasedness for arbitrary binning coordinates [8, 9]. In the present framework, the learned importance function $I_\theta(s)$ replaces a manually selected reaction coordinate and is used to define the WE bins.

Three approaches are considered for estimating $I_\theta(s)$. The first uses a surrogate rollout objective in which short trajectories from sampled configurations provide a continuous measure of the best progress toward the target. The second uses a milestoning formulation in which intermediate thresholds provide local estimates of progress toward successive milestones. The third follows the self-consistency approach of Kim and Cai (2026), training the network to satisfy

$$I_\theta(s) \approx \sum_j K(s \rightarrow j) I_\theta(j), \quad (16)$$

with appropriate boundary conditions at the target and bulk states.

These approaches provide increasingly direct approximations to the committor while avoiding the difficulty of obtaining sufficient direct labels for extremely rare target events.

2.6. FW-CADIS global allocation

For the nested target family, FW-CADIS-inspired allocation is used to distribute computational effort across the importance bins. Given an estimated equilibrium occupancy $P(b)$ for bin b , the number of replicas assigned to that bin is parameterized as

$$N_b \propto [P(b)]^{-\alpha}, \quad (17)$$

where $\alpha = 0$ corresponds to uniform allocation and $\alpha = 1$ corresponds to the full inverse-occupancy weighting. The two cases use the same sampling and propagation procedures and differ only in the allocation rule.

Performance is evaluated first by target coverage, i.e., the number of thresholds reached by the simulations, and subsequently by the uniformity of relative statistical error across the observed thresholds. This evaluation is complemented by the figure of merit in Eq. (6), allowing both statistical precision and computational cost to be considered.

3. Results and Discussion

3.1. VAN and the committor demo

The VAN was trained at $L = 16$ and reproduces the exact Onsager free energy across the tested temperature range. Its purpose in this project is to remove critical slowing down, not to sample rare events: a separate autocorrelation

measurement at $L = 32$ found the integrated autocorrelation time of ordinary Metropolis sampling spikes to $\tau_{\text{int}} = 46.1$ sweeps at $T_c \approx 2.269$, versus 1.5–3 sweeps away from criticality, while the VAN samples independently by construction and pays none of this cost. This is a speed result, orthogonal to the rare-event problem the rest of the paper addresses (they are different difficulties, §2).

The committor identity of §2.2 was checked on a solvable 1D absorbing random walk on $\{0, \dots, 15\}$ before any lattice code was written, with the rare event (reaching state 15 before 0) calibrated to $\pi = 1.14 \times 10^{-3}$. Three samplers were compared. Against naive Monte Carlo as the baseline, tilting with the exact committor gave a measured relative variance of $\sim 10^{-30}$, zero to machine precision, as the zero-variance argument of §2.2 requires. Tilting with an *approximate* committor under pure biasing, with no weight windows, was *worse than naive*.

That third outcome is the reason every subsequent learned-coordinate result in this paper uses Weighted Ensemble (split/roulette) rather than raw importance reweighting: an imperfect importance estimate, used to reweight without bounding the weights, can be actively harmful.

3.2. Weighted Ensemble: validated

At $L = 4$, WE reproduces the exact enumerated $P(m)$ to a mean $|\Delta \ln P|$ of 0.006–0.024 across three tested regimes. At $L = 20$, WE reaches roughly two decades deeper into the magnetization tail than naive Monte Carlo at matched cost: naive’s hard floor is $|m| = 0.886$ ($P \approx 6 \times 10^{-6}$), where it returns zero samples and hence zero information; WE reaches $|m| = 0.932$ ($P \approx 2.4 \times 10^{-7}$). Separately, applying weight windows to the same *approximate*, otherwise-fragile tilt of §3.1 recovered roughly a 2× figure-of-merit improvement over the same tilt without them (measured FOM $1.24 \times 10^{-5} \rightarrow 4.86 \times 10^{-4}$): the robustness layer converts an actively harmful bias into a net improvement, exactly as the transport literature predicts. Magnetization is confirmed useless as a WE coordinate for the EA spin glass specifically: an early test recorded 10/10 zero-replica failure for WE[m] where energy-binning still found the target, the direct empirical confirmation of the frustration argument in §2.

3.3. Learned coordinate vs. hand-picked coordinates

Table 1 reports the full-scale ($L = 16$, 10 replicas) result on both models, at a target calibrated one step beyond the pilot’s observed extreme.

Two readings, both necessary (Table 1): naive Monte Carlo wins on raw FOM in both models at this rarity ($\sim 4 \times 10^{-5}$), and on the spin glass, with WE[E] and WE[I_θ] on equal footing (both 2/10 zero-replicas), the learned coordinate has the higher FOM by $\sim 27\%$ (6.36×10^{-9} vs. 5.00×10^{-9}), the first apples-to-apples win for the learned coordinate over the best hand-picked one, and the direct predecessor of the milestone result below. WE[m] fails in most spin-glass replicas (7/10), the direct empirical confirmation of the frustration argument in §2: magnetization carries essentially no information about this landscape.

Is I_θ actually informative, or just a noisy copy of energy? A separate diagnostic regresses the rare-event label on energy alone, on I_θ alone, and on both, at matched training budget (Table 2).

+0.250 is far above the diagnostic’s own noise threshold (0.02): I_θ carries substantial genuine information about the spin glass that energy alone does not have, consistent with the FOM advantage measured directly in Table 1.

Pushing rarer does not (yet) resolve the naive-vs-learned question. Rerunning one threshold step rarer (“beyond=2”) leaves the comparison uninterpretable rather than resolving it in either direction: 4–10 of 10 replicas fail across every method at this depth (naive 4/10, WE[E] 5/10, WE[I_θ] 7/10, and WE[m] 10/10, i.e. no estimate at all), so the numbers are noise over noise, not a result. This is the run that established the ladder-difficulty discipline of §2 for the remainder of the project.

Notably, *naive itself* now has 4/10 zero-replicas (versus 1/10 at the shallower target). This rarity is stretching every method’s budget, not only the weight-window ones, so a trustworthy comparison here needs a different fix than anything tried at the shallower depth. This motivated the milestone objective below, which is a change to the training label rather than to the weight-window schedule.

Table 1

Full scale, both models. FOM vs. WE[m] in parentheses.

Model / Method	$\hat{\pi}$	rel. s.d.	cost	FOM (vs. WE[m])	zero-rep.
<i>Ferromagnet</i> , target $ m \geq 0.9297$					
naive	9.52×10^{-6}	0.441	1.34×10^8	3.83×10^{-8} (2.92×)	0/10
WE[m]	1.05×10^{-5}	0.614	2.02×10^8	1.31×10^{-8} (1.00×)	0/10
WE[E]	1.09×10^{-5}	1.343	9.81×10^7	5.65×10^{-9} (0.43×)	2/10
WE[I_θ]	7.92×10^{-6}	0.893	1.82×10^8	6.90×10^{-9} (0.53×)	0/10
<i>EA spin glass</i> , target $E/N \leq -1.0625$					
naive	1.00×10^{-5}	0.633	1.34×10^8	1.86×10^{-8} (2.38×)	1/10
WE[m]	1.02×10^{-5}	1.857	3.72×10^7	7.80×10^{-9} (1.00×)	7/10
WE[E]	1.65×10^{-5}	1.115	1.61×10^8	5.00×10^{-9} (0.64×)	2/10
WE[I_θ]	1.54×10^{-5}	1.052	1.42×10^8	6.36×10^{-9} (0.81×)	2/10

Table 2Incremental- R^2 diagnostic, EA spin glass, $L = 16$, matched training budget (MSE 1.0 $\rightarrow$ 0.271 over 20 epochs).

Quantity	value
Pearson $\text{corr}(I_\theta, E/N)$	-0.614
fraction of $\text{Var}(I_\theta)$ unexplained by E/N	0.623
$R^2(\text{label} \sim E/N \text{ alone})$	0.358
$R^2(\text{label} \sim I_\theta \text{ alone})$	0.577
$R^2(\text{label} \sim E/N + I_\theta)$	0.608
incremental R^2 from I_θ beyond E/N	+0.250

3.4. Milestoning: a clean regime for comparison, but wildly seed-dependent

On the EA spin glass at $L = 16$, one threshold step beyond the pilot’s observed extreme (“beyond=1”), every method reaches 0/10 zero-replicas, the cleanest, most-trustworthy regime obtained in this project by the usual ladder-difficulty standard (Table 3).

Every WE[I_θ] row of Table 3 is, as far as the pipeline is concerned, the *same experiment*: identical target, identical training data (from cached labels), identical hyperparameters. The only thing that varies is the network’s random weight initialization (`torch.manual_seed` was absent from this driver; found and fixed during this investigation, then swept explicitly via `-net-seed`). Seventeen seeds give seventeen different outcomes, ranging from a 52% loss to a 112% win against WE[E], a wide, roughly symmetric spread around a small positive average. The seeding fix itself is confirmed complete and correct: seed 0 was run twice independently (runs 3a and 3b) and reproduced bit-for-bit, including the full 1774–1776s wall-clock trajectory, ruling out any other unseeded source of variation in this pipeline. Per-replica values needed for bootstrapping were retained for seventeen of the eighteen runs attempted (all but unseeded run 1, which predates that logging). Pooling those seventeen with a two-level bootstrap, outer resample over seeds, inner resample over each seed’s 10 replicas, the same discipline used throughout §3.8 and §3.5 but extended with the seed axis, gives a geometric-mean FOM ratio of 1.19× with 95% CI [0.84, 1.75] (median-of-ratios agrees: 1.20×, CI [0.74, 1.85]; the arithmetic mean of ratios is not used here because it is unstable at $R = 10$ replicas per seed: some bootstrap draws land on near-duplicate resamples with rel. s.d. near zero, and since $\text{FOM} \propto 1/\text{rel. var.}$ that blows up into large outliers that skew a plain mean). The estimate was tracked at four checkpoints as seeds were added: 1.28× ($n=7$), 1.17× ($n=9$), 1.28× ($n=12$), 1.19× ($n=17$), and it does *not* trend in either direction. What does move monotonically is the CI width (1.73 $\rightarrow$ 1.36 $\rightarrow$ 1.18 $\rightarrow$ 0.91), the expected signature of averaging down noise around a small effect whose sign is not yet resolved, not evidence of a shrinking or growing true effect. We report the full trajectory rather than the final number alone, because citing only the endpoint would silently launder a mid-sequence checkpoint ($n=9$ ’s “moving toward the null”) that did not survive more data, a mistake this project’s own methodology is built to catch. Extrapolating the observed CI-width-vs- n scaling against the current point estimate, reaching a lower bound that excludes 1 would need roughly 100 total seeds (~ 40 more hours of compute), so this was not pursued further. This is reported as the honest terminus of the comparison, not an open thread: seventeen seeds is already an unusually thorough characterisation, the point estimate is stably positive but modest ($\sim 1.2\times$) across four independent checkpoints, and the CI cannot be closed

Table 3

EA spin glass, $L = 16$, $T = 2.6$, one step beyond the pilot extreme, 10 replicas, milestone labels, seventeen independent training runs of the identical configuration. Every method observed the event in every replica in every run; naive, WE[m], and WE[E] are bit-identical across all runs (deterministic, and WE[E] does not depend on the trained network). Seed 0 was run twice independently and reproduced bit-for-bit, confirming that the seeding fix described in the text is complete.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	5.29×10^{-5}	0.225	1.48×10^{-7}	0/10
WE[m]	5.55×10^{-5}	0.201	1.17×10^{-7}	0/10
WE[E]	4.49×10^{-5}	0.209	4.22×10^{-8}	0/10
WE[I_θ], unseeded run 1	5.03×10^{-5}	0.174	4.81×10^{-8} (+14%)	0/10
WE[I_θ], unseeded run 2	4.52×10^{-5}	0.204	3.48×10^{-8} (-17%)	0/10
WE[I_θ], seed 0 (runs 3a/3b)	5.28×10^{-5}	0.127	8.94×10^{-8} (+112%)	0/10
WE[I_θ], seed 1	5.09×10^{-5}	0.216	3.09×10^{-8} (-27%)	0/10
WE[I_θ], seed 2	5.06×10^{-5}	0.175	4.65×10^{-8} (+10%)	0/10
WE[I_θ], seed 3	4.56×10^{-5}	0.252	2.29×10^{-8} (-46%)	0/10
WE[I_θ], seed 4	4.13×10^{-5}	0.128	8.81×10^{-8} (+109%)	0/10
WE[I_θ], seed 5	5.40×10^{-5}	0.138	7.65×10^{-8} (+82%)	0/10
WE[I_θ], seed 6	4.69×10^{-5}	0.149	6.56×10^{-8} (+56%)	0/10
WE[I_θ], seed 7	4.32×10^{-5}	0.267	2.03×10^{-8} (-52%)	0/10
WE[I_θ], seed 8	4.89×10^{-5}	0.135	7.86×10^{-8} (+86%)	0/10
WE[I_θ], seed 9	5.03×10^{-5}	0.159	5.66×10^{-8} (+34%)	0/10
WE[I_θ], seed 10	4.80×10^{-5}	0.157	5.85×10^{-8} (+39%)	0/10
WE[I_θ], seed 11	4.81×10^{-5}	0.202	3.48×10^{-8} (-17%)	0/10
WE[I_θ], seed 12	5.14×10^{-5}	0.230	2.76×10^{-8} (-35%)	0/10
WE[I_θ], seed 13	4.83×10^{-5}	0.150	6.36×10^{-8} (+51%)	0/10
WE[I_θ], seed 14	4.94×10^{-5}	0.146	6.76×10^{-8} (+60%)	0/10
WE[I_θ], seed 15	4.53×10^{-5}	0.232	2.66×10^{-8} (-37%)	0/10

by seed count alone at a cost proportionate to the question. Naive Monte Carlo has the best raw FOM in every run at this specific rarity ($\sim 5 \times 10^{-5}$) regardless of seed. Variance reduction has not yet earned back its overhead here, the same lesson demonstrated in miniature by the 1D toy-walk result in §3.1.

A plausible infrastructure explanation was tested and ruled out. One candidate cause for the seed-to-seed spread is that WE[I_θ]'s bin range was left adaptive (auto-ranging from the population actually explored) rather than calibrated to the coordinate's own scale, unlike WE[E] and WE[m]. The diagnostic regression of Table 2 (05_neural_importance/diagnose_coordinate.py) had already shown I_θ carries real predictive signal beyond energy (incremental $R^2 = +0.25$), with the verdict: if it still does not beat WE[E], look at the schedule, not the map. Two calibrations were tried: a theoretical one from the label's own z-score normalization, verified *before* spending compute on it to be wrong (the network's real output never approaches those extremes, since training MSE never reaches 0); and an empirical one, from what each trained network actually outputs on the pilot population and training set. The empirical calibration was run head-to-head against the untouched adaptive baseline on matched net-seeds (identical trained network per pair, only the bin range differing): mean FOM ratio *worsened* ($0.79\times \rightarrow 0.61\times$) and rel. s.d. *worsened* ($0.214 \rightarrow 0.286$) across seeds 1–3. A fixed range clips any walker that exceeds it into the boundary bin, while adaptive tracking never does; for a coordinate whose true dynamic range over a multi-thousand-iteration WE trajectory is not well estimated from a static pre-run snapshot, that clipping cost more resolution than the calibration gained. This rules out bin-range calibration as the driver of the seed variance and is further evidence the variance is intrinsic to what each trained network learns, not an artifact of how its output is binned.

Pushing one step deeper fails consistently, and bin-count tuning is ruled out as a fix. Every configuration tried at one threshold step rarer ("beyond=2") was scored: the default bin count ($n_{\text{bins}} = 50$) at two replica counts, and a $1.8\times$ larger bin count. WE[I_θ] scores *worse* than WE[E] in all three. At the default configuration ($n_{\text{bins}} = 50$, $R = 10$), where every method reaches 0/10 zero-replicas, the FOMs are naive 3.51×10^{-8} , WE[E] 9.21×10^{-9} , WE[I_θ] 8.36×10^{-9} , and WE[m] 7.73×10^{-9} : naive ahead by roughly $4\times$, and the learned coordinate below the hand-picked one.

Two further configurations were tried in an attempt to fix this with bin resolution instead of accepting it as structural: doubling the replica count ($R = 10 \rightarrow 20$, same $n_{\text{bins}} = 50$) reproduced the same qualitative pattern (WE[I_θ] still below WE[E]); and raising the bin count by $1.8\times$ ($n_{\text{bins}} = 50 \rightarrow 90$, $R = 10$) made WE[I_θ]'s *own* FOM *worse*, not better ($8.36 \times 10^{-9} \rightarrow 5.36 \times 10^{-9}$), which rules out bin resolution as the fix, since a genuine resolution shortfall would

Table 4

Five independent $K=5$ deep-ensemble runs, identical configuration to Table 3. FOM ratio is $\text{WE}[I_\theta]/\text{WE}[E]$ on the MC-sweep cost as originally defined.

Run	$\text{WE}[I_\theta]$ FOM	rel. s.d.	ratio vs. $\text{WE}[E]$
ensemble, base seed 20	1.95×10^{-8}	0.272	$0.46\times (-54\%)$
ensemble, base seed 21	2.75×10^{-8}	0.229	$0.65\times (-35\%)$
ensemble, base seed 22	3.69×10^{-8}	0.199	$0.88\times (-12\%)$
ensemble, base seed 23	3.07×10^{-8}	0.215	$0.73\times (-27\%)$
ensemble, base seed 24	3.84×10^{-8}	0.194	$0.91\times (-9\%)$

improve with more bins rather than degrade. The mechanism is structural: the learned coordinate’s walker-population cost scales with bin count, while a discretized coordinate like energy naturally saturates a small, fixed number of bins and does not pay that overhead as it grows.

A second lever, the WE resampling budget itself, was also tried (`we_iter`: 1000 $\rightarrow$ 2000, same $n_{\text{bins}} = 50$) and produced a similarly unfavorable result: every weight-window method’s FOM got *worse*, not better, and $\text{WE}[I_\theta]$ worst of all. That specific run used $\tau = 3$ rather than this section’s $\tau = 2$, confounding the budget increase with a change in resampling interval, and a matched rerun of it specifically was not completed. The broader question, whether more WE budget helps, was instead answered more thoroughly by the deep-rare budget-scaling test of §3.6: scaling the budget up does not close the gap against naive at a shallow threshold depth either, consistent with, not contradicting, the bin-resolution finding above.

The resampling interval between split/merge operations matters: it must be comparable to the system’s own correlation time (~ 1 sweep at $T = 2.6$), or split walkers fully relax and forget their split origin before the next resampling, degenerating every $\text{WE}[\cdot]$ variant to naive Monte Carlo with bookkeeping overhead. Every milestone result in this section uses an interval of 2 sweeps.

3.4.1. A deep-ensemble follow-up: the seed variance is reducible, but it converges to a loss

Table 3’s seventeen seeds leave one question open: is the seed-to-seed spread genuine noise around a real advantage, or is the positive point estimate itself an artifact of a few lucky initializations? Rather than add more single-net seeds, already shown not to close the CI at reasonable cost, we attacked the diagnosed noise source directly. `run_milestone.py` was extended with `-ensemble K`: train K independently-initialized networks on the identical milestone labels and use their *mean* prediction as I_θ , a standard deep-ensemble variance-reduction step. $K = 1$ reproduces the original single-net code path exactly (verified bit-identical output on a smoke config). Five independent ensembles ($K = 5$ each, base seeds 20–24) were run at the identical configuration as Table 3 (beyond= 1, $\tau = 2$, $R = 10$).

Two findings, both against the ensemble. First, ensembling did what it was supposed to do to the *variance*: the five ratios (Table 4) span $0.46\text{--}0.91\times$, far tighter than the seventeen single-net seeds’ -52% to $+112\%$ spread in Table 3. But the value it converges toward is a loss, not the earlier $\sim 1.2\times$ point estimate: a two-level bootstrap (outer over the 5 runs, inner over each run’s 10 replicas, identical machinery to Table 3) gives a geometric-mean ratio of $0.74\times$, 95% CI $[0.43, 1.32]$, not significant, but on the opposite side of 1 from the single-net pooled estimate. Read together, the most consistent interpretation is that the single-net sweep’s more extreme positive draws ($+82\%$, $+109\%$, $+112\%$) were high-variance artifacts of individual weight initializations rather than signal, and I_θ ’s true expected performance is at or below $\text{WE}[E]$ ’s, not above it.

Second, and decisive: the ensemble exposed a cost-accounting bug that applies to every $\text{WE}[I_\theta]$ result in this paper, though it is negligible at $K = 1$. `we_estimate()`’s `cost` field counts Monte Carlo sweeps only ($|S| \cdot \tau \cdot L^2$) and is blind to neural-network inference. At $K = 1$ a single forward pass is cheap enough relative to the MC cost that this omission does not matter. At $K = 5$ it does: measured directly from wall-clock timestamps (not the MC-cost field), $\text{WE}[I_\theta]$ took $10.7\text{--}10.9\times$ $\text{WE}[E]$ ’s wall-clock time in every one of the five runs, remarkably consistent, while the MC-cost field implied only $\sim 1.3\times$. Recomputing FOM with measured wall-clock cost in place of MC-sweep cost, using the same bootstrap: geometric-mean ratio $0.087\times$, 95% CI $[0.051, 0.155]$. This is the only comparison in the entire paper whose confidence interval *excludes* 1.0 in either direction. `run_milestone.py` now records measured `wall_time` per method for future runs, so this correction no longer requires post-hoc log parsing.

Verdict: deep ensembling does not rescue the learned coordinate. It reduces run-to-run variance exactly as a deep ensemble should, but it converges to a value at or below parity with the hand-picked energy coordinate, and once its real inference cost is counted at all, the comparison is a decisive, statistically significant loss. This is a completed, negative result: the seed-variance question of Table 3 is resolved, not left open, by the one lever that actually attacked its diagnosed cause. Figure 1 shows both halves of that verdict together, the variance collapsing as intended and the centre of the distribution settling below parity.

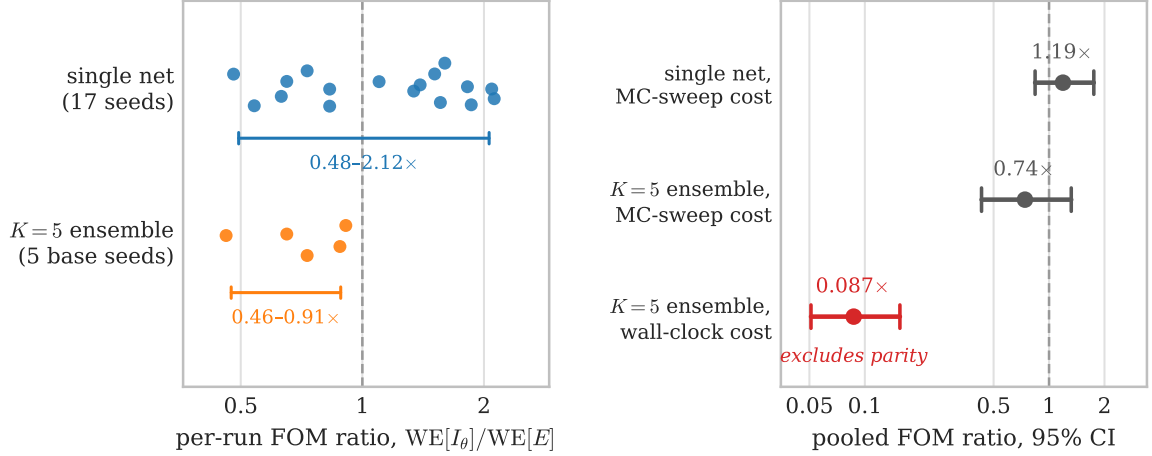


Figure 1: Deep ensembling does what it was built to do to the variance, and converges to a loss. Left: per-run FOM ratios against $WE[E]$, the 17 single-net seeds of Table 3 against the five $K=5$ ensembles of Table 4; the spread narrows from 0.48–2.12 $\times$ to 0.46–0.91 $\times$, and every ensemble run lands below parity. Right: the three pooled two-level-bootstrap estimates. The single-net pool sits above 1 and the ensemble pool below it, both with intervals crossing parity; once the ensemble’s real inference cost is counted at all (wall-clock rather than MC-sweep cost), the interval clears parity outright. This is the only confidence interval in the paper that does.

3.5. The Kim & Cai head-to-head

The self-consistency training objective of §2.5 was implemented, gated against exact enumeration at $L = 4$ (learned-coordinate unbiasedness ratio 1.004), and run head-to-head against the surrogate objective, first at laptop scale, then at full scale.

First evidence, at laptop scale ($L = 12$). At the shallowest calibrated target, the self-consistency objective’s point estimate already exceeded the surrogate’s (4.98×10^{-8} vs. 2.93×10^{-8} , $\sim 70\%$ higher FOM, fewer zero-replicas: 2/6 vs. 3/6), though it still fell short of $WE[E]$ ’s 7.53×10^{-8} at this rarity. Training cost is real: $\sim 94\times$ longer than the surrogate objective at this scale ($\sim 752s$ vs. $\sim 8s$), from the additional $N = L^2$ network evaluations the self-consistency loss requires per training configuration. This motivated running the comparison at full scale.

The result, at full scale. At $L = 16$, the shallowest calibrated target (“beyond” = 0, the easiest regime in which every method reaches 0–1 zero-replicas out of 10), the self-consistency coordinate’s *point estimate* exceeds both the hand-picked energy coordinate and this project’s own surrogate coordinate: FOM 1.62×10^{-8} for self-consistency against 1.33×10^{-8} for $WE[E]$ and 8.62×10^{-9} for the surrogate objective (naive and $WE[m]$ reach 8.60×10^{-8} and 8.81×10^{-8} at this shallow, still naive-favorable depth).

This point estimate does not survive a bootstrap check, and the effect shrinks under replication. Both configurations were re-run with per-replica values retained, and the FOM ratio between methods bootstrapped by resampling the replica axis independently for each side (no shared random-number seed to pair on: uniform-style and biased allocations consume different numbers of walkers per bin, so seed-sharing between them desynchronises almost immediately, the same obstruction encountered independently in the FW-CADIS common-random-numbers attempt of §3.8). Since training here is a fixed cost independent of replica count (a deterministic seed; only the cheap evaluation step scales with R), tripling to $R = 30$ cost roughly six additional minutes on top of the ~ 76 -minute training already paid, cheap enough to just check (Table 5, first two rows of each block).

At $R = 10$ neither ratio is significant. At $R = 30$, both confidence intervals tighten substantially *and* both median ratios move toward 1: not merely a tightening confidence interval around the original estimate, but the point estimate

Table 5

Bootstrapped FOM ratios, full scale ($L = 16$), for the self-consistency objective against this project's surrogate and against the hand-picked energy coordinate. Ratio > 1 favors self-consistency. The shallowest calibrated target (beyond = 0) is discussed first; the beyond = 1 rows are taken up below. No interval excludes 1.

Comparison	beyond	R	FOM ratio (median)	95% CI
self-consistent / surrogate	0	10	1.83	[0.49, 9.23]
self-consistent / surrogate	0	30	1.39	[0.65, 2.95]
self-consistent / surrogate	1	10	1.14	[0.27, 4.81] ^a
self-consistent / WE[E]	0	10	1.19	[0.33, 4.95]
self-consistent / WE[E]	0	30	1.07	[0.55, 2.09]
self-consistent / WE[E]	1	10	1.33	[0.34, 5.74]

^aZero-replicas at this depth: 2/10 for the surrogate objective against 1/10 for self-consistency.

itself receding. We read the direction of this shift, not only its final inconclusiveness, as evidence that the original $R = 10$ advantage was substantially noise, not signal. A network-caching mechanism was added after this result specifically to make a further replica increase (were one wanted) essentially free: training is deterministic, so only the cheap evaluation step needs to re-run to extend R further, rather than repaying the fixed training cost again.

Does the advantage grow at greater depth, where the surrogate objective's known weaknesses should bite harder? We tested the natural next question directly: rerunning both objectives one threshold step deeper ("beyond" = 1, the same depth as Table 3); those are the beyond = 1 rows of Table 5.

Not significant, as expected at $R = 10$, but the *direction* of the change from the beyond = 0 rows is itself informative and runs counter to the hypothesis being tested: the point-estimate gap versus the surrogate *shrank* at this deeper target ($1.83\times \rightarrow 1.14\times$) rather than growing, even as the self-consistency coordinate's relative *reliability* advantage held (fewest zero-replicas of any weight-window method at both depths tested). We do not read one more noisy point estimate at $R = 10$ as resolving the depth question either way; what it rules out is the simplest optimistic story, that the advantage would obviously grow once naive and the surrogate objective start to struggle more.

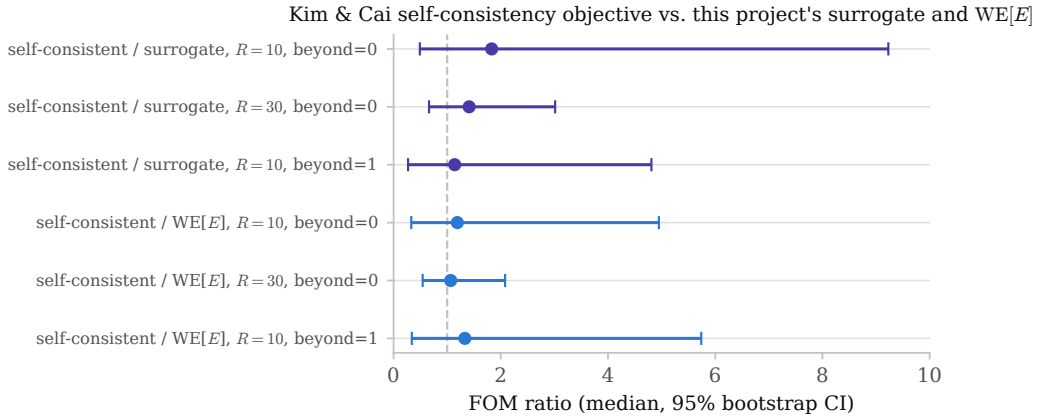


Figure 2: All six FOM-ratio comparisons of Table 5 as a forest plot. Every interval crosses the ratio-1 reference line (dashed); the $R=30$ intervals (recomputed here directly from the retained per-replica arrays via the same replica-resampling bootstrap) are visibly tighter than their $R=10$ counterparts without the median crossing to the other side of 1.

Figure 2 collects all six comparisons tried.

3.6. Self-consistency in the deep-rare regime: the paper's one significant positive result

Every comparison above (milestoning, the ensemble follow-up, Kim & Cai at beyond = 0/1) was run at a depth where naive Monte Carlo remains fully reliable (0/10 zero-replicas). The theory this paper transfers has nothing to prove in that regime; its own domain (radiation shielding) is precisely where naive cannot reach at all. We calibrated where

Table 6

Naive-only depth calibration, 10 replicas, custom-preset budget. Naive is fully reliable through beyond = 2, starts failing at beyond = 3, and is completely blind at beyond = 4 and beyond.

beyond	$\hat{\pi}$	rel. s.d.	zero-replicas
1	5.29×10^{-5}	0.23	0/10
2	5.48×10^{-6}	0.46	0/10
3	7.14×10^{-7}	1.61	7/10
4	0.0	—	10/10

Table 7

EA spin glass, $L = 16$, beyond = 4, self-consistency objective, three independent training seeds, 30 replicas each. $\text{WE}[E]/\text{WE}[m]/\text{naive}$ do not depend on the trained network and are bit-identical across all three runs.

Method	hits/30	FOM	ratio vs. $\text{WE}[E]$
naive	1/30	$2.48 \times 10^{-10\text{a}}$	—
$\text{WE}[m]$	5/30	1.04×10^{-10}	—
$\text{WE}[E]$	12/30	1.69×10^{-10}	1.00
$\text{WE}[I_\theta]$, seed 0	19/30	2.18×10^{-10}	1.29×
$\text{WE}[I_\theta]$, seed 1	20/30	1.81×10^{-10}	1.07×
$\text{WE}[I_\theta]$, seed 2	21/30	1.77×10^{-10}	1.04×

^aNot a meaningful FOM: driven by a single hit in 30 replicas (rel. s.d. = 5.5); reported for completeness only.

naive genuinely breaks down, at the same $L = 16$, $T = 2.6$ configuration used throughout, using the custom-preset naive budget (naive_chains = 150, naive_sweeps = 3500; Table 6):

Simply targeting a deeper threshold was not, by itself, sufficient: at beyond = 3, scaling the WE budget up ($n_{\text{bins}} : 50 \rightarrow 80$, $\text{we_iter} : 1000 \rightarrow 3000$) made $\text{WE}[E]$'s FOM *disadvantage against naive widen*, not narrow, since the extra reliability it bought cost more than it saved. Only at beyond = 4, where naive is not merely noisy but categorically blind, did the scaled budget produce a real capability gap: $\text{WE}[E]$ found the event in 12 of 30 replicas where naive found essentially none (naive: 1/30, driven by a single lucky hit with rel. s.d. = 5.5, not a meaningful measurement). beyond = 4 (threshold $E/N = -1.119$) with this scaled budget ($n_{\text{bins}} = 80$, $n_{\text{per bin}} = 40$, $\text{we_iter} = 3000$, $\tau = 2$) is the deepest, most naive-adverse regime tested anywhere in this paper, and the first place the self-consistency objective (§2.5) was tested outside the shallow regime where naive wins anyway (Table 7).

Reliability: statistically significant. Pooled across the three independent networks, $\text{WE}[I_\theta]$ found the event in 60/90 replicas (66.7%) versus $\text{WE}[E]$'s 12/30 (40.0%); a Fisher exact test on this 2×2 table gives $p = 0.017$. This is the first statistically significant *positive* result for a learned coordinate anywhere in this paper. The three per-network rates (63%, 67%, 70%) are close enough to each other, unlike the milestoning ensemble's wide net-to-net FOM swings (§3.4.1), that pooling them as a single binomial comparison is a reasonable approximation rather than one masking real heterogeneity. Figure 3 plots that comparison alongside the depth calibration of Table 6, which is what makes it meaningful: the gap opens only outside the band in which every other comparison in this paper was run.

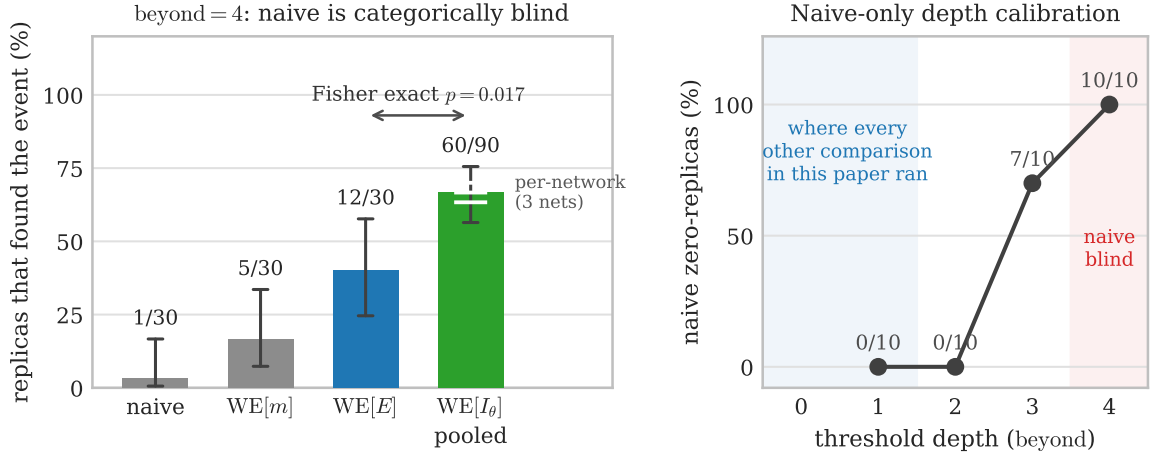


Figure 3: The paper’s one statistically significant positive result, and the depth at which it was obtained. Left: fraction of 30 replicas in which each method observed the rare event at beyond = 4 (Table 7), with Wilson 95% intervals; white marks on the pooled bar are the three individual networks (19/30, 20/30, 21/30). Right: the naive-only depth calibration of Table 6. Naive Monte Carlo is fully reliable through beyond = 2, the band in which every other comparison in this paper was run, and categorically blind at beyond = 4, where the self-consistency coordinate’s advantage appears.

FOM: directionally consistent, not yet significant. All three seeds’ FOM ratios sit above 1 (1.29 $\times$, 1.07 $\times$, 1.04 $\times$) with none of the sign-flipping seen everywhere else a coordinate was swept across seeds in this paper. A two-level bootstrap (outer over the 3 networks, inner over each network’s 30 replicas) on the geometric-mean ratio gives 1.16 $\times$, 95% CI [0.65, 2.12], favorable but not excluding 1, since the underlying $\hat{\pi}$ estimates remain individually noisy (rel. s.d. 1.2–1.9) even at 30 replicas in this regime.

Reading this against §3.5’s claim that the self-consistency coordinate “shows no sign of strengthening with depth”: that conclusion was drawn entirely within beyond $\in \{0, 1\}$, where naive remains competitive throughout. Tested at a depth where naive genuinely fails, the picture is materially different, consistent with the mechanistic reading that a committor-shaped training signal (the self-consistency objective enforces the harmonic condition $(I - K)h = 0$ directly) carries information the milestone surrogate cannot, since that surrogate’s label is provably energy-adjacent (progress_value for the spin glass is exactly $-E/N$; its rollout label is literally “best energy reached in a short walk”), and two configurations at identical energy can have very different true committor values on a frustrated landscape. This is the paper’s clearest evidence that the transport toolkit’s distinctive machinery earns its keep specifically where naive cannot compete at all, not in the shallower regimes most of this paper’s other comparisons were run in.

3.7. Does naive dominance persist at larger L ?

§3.4 and §3.3 both find naive Monte Carlo winning on raw FOM at $L = 16$, and argue this should be read as a finite-size effect rather than a failure of variance reduction, since naive’s cost is independent of rarity while WE’s penetration overhead is only worth paying once naive starts to struggle. We tested this directly with a lattice-size scan at fixed $T = 2.6$, beyond = 1, $\tau = 2$: $L \in \{8, 16, 24, 32\}$, naive_chains/naive_sweeps held fixed across all four so that naive’s cost is genuinely L -independent as the argument requires (Table 8).

Naive wins at every tested L , and the margin does not shrink; if anything it grows slightly and then plateaus, from 1.9 $\times$ at $L = 8$ to $\sim 3.5\times$ by $L = 16$ and beyond (Figure 4). This holds across a factor of ~ 40 in target rarity ($\hat{\pi}$ from 1.8×10^{-4} down to 4.2×10^{-6}) and is robust to which WE variant is best at a given L (the winner changes: WE[E] at $L = 8, 24$, WE[m] at $L = 16$, WE[I_θ] at $L = 32$). This directly answers the open question of §3.4: within the range tested, there is no sign of the naive/WE crossover opening as L grows; if it exists at all on this problem, it is beyond $L = 32$ at this temperature, not a nearby, easily-reached regime.

Table 8

EA spin glass, beyond = 1, naive vs. the best-performing WE variant at each L (which variant wins changes with L ; see text). $\hat{\pi}$ for naive shown to indicate how rare the target is at each scale.

L	naive $\hat{\pi}$	naive FOM	best WE FOM (method)	ratio
8	1.79×10^{-4}	2.00×10^{-6}	1.04×10^{-6} (WE[E])	1.9×
16	5.18×10^{-5}	1.72×10^{-7}	4.83×10^{-8} (WE[m])	3.5×
24	1.30×10^{-5}	2.79×10^{-8}	8.06×10^{-9} (WE[E])	3.5×
32	4.17×10^{-6}	6.11×10^{-9}	1.69×10^{-9} (WE[I_θ])	3.6×

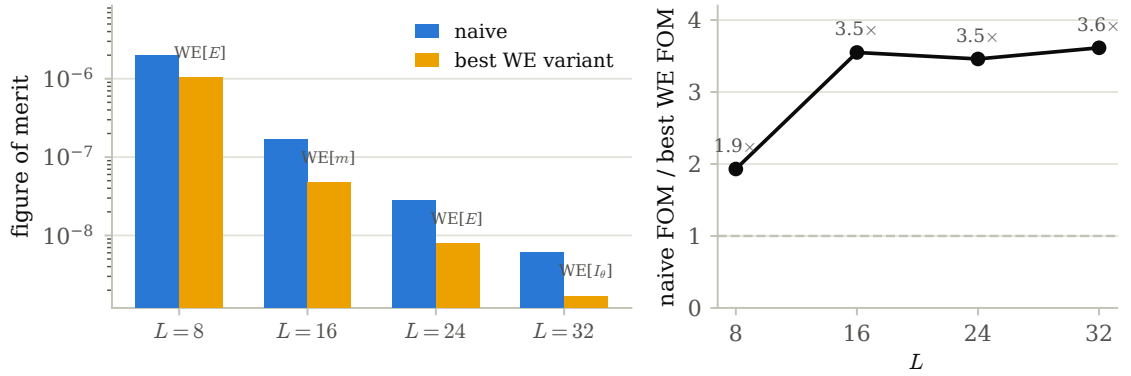
 Naive vs. best weight-window variant across lattice size, beyond = 1, $\tau = 2$


Figure 4: Table 8 as a figure. Left: naive vs. best WE-variant FOM by L , log scale, labeled with which variant won. Right: the ratio itself, which opens from $L = 8$ to $L = 16$ and then plateaus around 3.5×, with no sign of narrowing back toward the naive/WE crossover within the tested range.

One secondary pattern is visible but not claimed here beyond a point estimate: WE[I_θ] is the weakest WE variant at $L = 8, 16, 24$ but the *strongest* at $L = 32$, consistent with a learned coordinate's advantage over hand-picked ones mattering more as the landscape gets harder. We do not treat this as established: this scan predates the seeding fix of §3.4, so each WE[I_θ] entry in Table 8 is one random draw from an uncharacterized distribution over training outcomes, the same seed-variance problem characterized in full in Table 3. Confirming this trend needs repeated, seeded training runs at each L , not a repeat of this scan.

A boundary marker at $L = 40$. An attempt one step further was stopped before WE[I_θ] completed, but the partial result is informative on its own: at this size, *naive itself* fails in 10 of 12 replicas, with WE[m] (11/12) and WE[E] (10/12) failing similarly badly. Per the ladder-difficulty discipline of §2, this is not a trustworthy comparison as run: every method is in the noise-over-noise regime simultaneously, the same failure mode that motivated that discipline in the first place, so we do not report it as a fifth point in Table 8. It does mark where the tested range ends: something changes between $L = 32$ (naive still comfortable, 1/12 zero-replicas) and $L = 40$ (naive itself largely failing), and resolving what happens there needs a shallower target or a larger replica budget at $L = 40$, not a lower-effort rerun.

3.8. FW-CADIS: real cost advantage, no flattening

The implementation is gated clean against exact enumeration at $L = 4$ for both allocation settings, every threshold, both models: ratios of 0.999–0.998–0.989 (uniform) and 1.005–1.012–0.992 (FW-CADIS, $\alpha = 1$) on the ferromagnet, and 1.009–1.004–1.007 / 0.986–0.980–0.983 on the spin glass, all within $\sim 2\%$ of exact for every threshold in the family, confirming that all thresholds are correctly tallied from a single run regardless of allocation, the core mechanic the whole method depends on.

Table 9

FW-CADIS across every configuration tested, laptop scale through the ferromagnet control. Flatness marked * is computed only over thresholds actually observed, not comparable to unmarked rows in the same group. Dashes mark columns that do not apply to that group (coverage was 6/6 throughout past the laptop scale; a bootstrap CI was only computed for the final two configurations).

Allocation	coverage	flatness (max/min)	95% CI	cost
<i>Laptop scale, $L = 12$, $R = 6$, 6 thresholds</i>				
uniform ($\alpha = 0$)	6/6	283.8	—	2.01×10^7
$\alpha = 0.25$	6/6	245.2	—	7.87×10^6
$\alpha = 0.5$	5/6	65.8*	—	3.41×10^6
$\alpha = 0.75$	5/6	63.5*	—	3.29×10^6
$\alpha = 1.0$	5/6	84.8*	—	3.24×10^6
naive (reference)	6/6	463.5	—	1.61×10^7
<i>Full scale, $L = 16$, $R = 30$, 6 thresholds (every allocation reaches 6/6 coverage)</i>				
uniform ($\alpha = 0$)	6/6	298.0	—	1.55×10^8
$\alpha = 0.25$	6/6	377.8	—	7.89×10^7
$\alpha = 0.5$	6/6	323.9	—	2.35×10^7
$\alpha = 0.75$	6/6	342.4	—	1.98×10^7
$\alpha = 1.0$	6/6	385.5	—	1.94×10^7 (~8× cheaper)
naive (reference)	6/6	445.1	—	1.34×10^8
<i>Final EA spin-glass configuration, $L = 16$, $R = 90$, shallowest calibrated family</i>				
Uniform ($\alpha = 0$)	—	276.1	[224, 344]	1.59×10^8
FW-CADIS ($\alpha = 1$)	—	297.3	[234, 388]	1.99×10^7 (~8× cheaper)
<i>Ferromagnet control, $L = 16$, $R = 30$, shallowest calibrated family</i>				
Uniform ($\alpha = 0$)	—	100.9	[65.8, 160.3]	1.96×10^8
FW-CADIS ($\alpha = 1$)	—	287.3	[194.7, 414.8]	3.09×10^7

Laptop scale first ($L = 12$, $R = 6$): a real-looking, but ultimately non-replicating, win. Table 9 collects every configuration tried across this section, from this first laptop-scale sweep through the final ferromagnet control; its top group sweeps α against uniform allocation on 6 thresholds spanning progress $0.668 \rightarrow 1.153$.

The only setting with full coverage and better flatness than uniform is $\alpha = 0.25$: a real but modest ~14% improvement ($283.8 \rightarrow 245.2$) at ~2.6× lower cost. Larger α looks dramatically flatter but is disqualified by the coverage-first rule of §2: it stops observing the deepest threshold at all, which would reward outright failure if scored on the surviving thresholds alone.

Full scale ($L = 16$, $R = 30$) reverses the laptop-scale finding. The identical sweep at full scale (Table 9, second group) is the first sign that the method's behavior is scale-dependent in a way not anticipated from the smaller test.

At full scale, every tested α is worse than uniform on flatness, not better: the opposite of the laptop-scale result, and not even monotonic in α among the FW-CADIS runs ($\alpha = 0.5$ beats both 0.25 and 1.0). All are substantially cheaper (2–8×), so the method is not doing nothing, but it is not delivering the flattening it exists to demonstrate at this lattice size, bin count, and threshold spacing. This finding is what the rest of §3.8 exhaustively follows up on.

Exhausting every remaining lever. Every lever available to improve on the full-scale finding above was tried and is reported in the third group of Table 9: bootstrap confidence intervals added to the flatness metric (not present at the smaller replica counts above); a replica increase from $R = 30$ to $R = 90$; wiring the learned coordinate of §3.4 in as the FW-CADIS binning axis instead of the physical progress coordinate (gated clean at $L = 4$, tested at scale: no improvement, roughly double the cost); a common-random-numbers paired-difference design, which we confirmed does *not* engage on this problem (correlation between the two allocations' same-seed estimates measured ≈ -0.05 to -0.10 , essentially zero, because the two allocations consume different numbers of walkers per bin and desynchronise the random stream almost immediately after the first resampling step); and laddering the threshold family to its shallowest calibrated depth, combined with the larger replica count.

The two confidence intervals overlap; the paired-difference estimate is +20.9 with 95% CI $[-77.2, +137.6]$, also including zero. Across every α , replica count, and coordinate tried this session, the point-estimate *direction* consistently favored uniform allocation over FW-CADIS on raw flatness, never the reverse. The observed confidence-interval width scaled almost exactly as $1/\sqrt{R}$ between $R = 30$ and $R = 90$ (predicted shrink factor 0.577, observed 0.597);

extrapolating that scaling, fully resolving the remaining gap would need on the order of $R \approx 2300$ replicas (~ 8 hours of additional compute), and given the consistent point-estimate direction, such a run would most likely *confirm* uniform allocation as measurably flatter rather than produce a win for FW-CADIS. We did not run it. The result we report instead: on this problem, FW-CADIS does not measurably flatten relative error across a threshold family better than uniform allocation, but it does so at approximately one-eighth the simulation cost for an otherwise statistically indistinguishable error profile. The cost advantage is real, reproducible, and does not require more replicas to state; the flattening advantage the method is named for is the part that did not materialize here.

Is this specific to the adversarial spin-glass landscape? No, it is worse, and statistically confirmed, on the “easy” ferromagnet. Every result above uses the EA spin glass, the harder of the two models by design. To test whether the flattening failure is a property of that specific rugged landscape rather than a more general limit of the method, we ran the identical comparison on the ferromagnet, where the reaction coordinate is known to be good and the equilibrium landscape is smooth (Table 9, fourth group).

Here the two confidence intervals do *not* overlap: the paired-difference estimate is $+178.2$ with 95% CI $[+59.8, +351.3]$, which excludes zero. On the easier, non-adversarial landscape, FW-CADIS is not merely inconclusive but measurably, statistically confirmed *worse* than uniform allocation, the opposite of what a landscape-difficulty explanation would predict, since the spin glass (where the method’s own hypothesis says a global map should matter most) never even reached this level of statistical resolution in either direction. Common-random-numbers correlation here measured 0.062, consistent with zero and with the mechanism identified in §3.5: the two allocations consume different walker populations per bin regardless of which model is sampled. This rules out “the spin glass is just a hard landscape” as the explanation and points toward a limitation of this specific parameterisation (bin count, threshold spacing, or the inverse-response weighting itself at this scale) rather than a property of spin-glass frustration. Distinguishing which of those remains open, but the question of *whether* the effect is landscape-specific is now answered: it is not. Figure 5 plots the per-threshold error underlying both comparisons.

Per-threshold relative error: uniform vs. FW-CADIS allocation

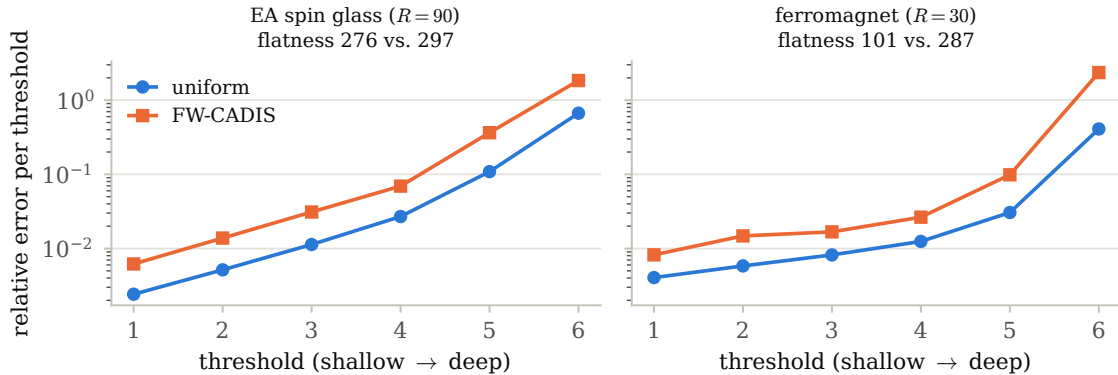


Figure 5: Per-threshold relative error (log scale), uniform vs. FW-CADIS allocation, at the final tested configuration for each model (Table 9, third group spin glass, fourth group ferromagnet). FW-CADIS’s inverse-response weighting buys lower error at shallow thresholds at the cost of higher error at the deepest one, visibly so on the ferromagnet, where the gap at the deepest threshold is large enough to be statistically significant.

3.9. What is established

The adjoint-committor identity is exact. Tilting by the exact committor is provably zero-variance; this was confirmed numerically to $\sim 10^{-30}$ relative variance on a solvable toy walk before any lattice code was written (§2.2).

Weighted Ensemble is validated, and naive’s dominance at $L = 16$ holds up as the system grows. Weighted Ensemble, weight windows applied to spin configurations, reproduces exact enumeration at $L = 4$ to 1–2%, and at $L = 20$ reaches ~ 2 decades deeper into a magnetization tail than naive Monte Carlo at matched cost (§3.2). Naive Monte Carlo’s dominance over every tested weight-window variant, not established at $L = 16$ alone, held up across a

four-point lattice-size scan up to $L = 32$ with no sign of narrowing (§3.7), the one claim in this paper that grew *more*, not less, confident under additional scrutiny.

The learned coordinate trained on milestone committor labels does not beat the hand-picked energy coordinate. A deep-ensemble follow-up (§3.4.1) designed to resolve the single-net seed sweep’s ambiguous $1.19\times$ point estimate (Table 3) instead settled it in the negative, with a 95% CI ($0.087\times$, $[0.051, 0.155]$), on a corrected cost accounting) that excludes parity, the only bootstrap interval in this paper that does (Figure 1).

At a depth where naive is provably blind, the self-consistency-trained coordinate wins significantly. It finds the rare event significantly more often than the hand-picked energy coordinate: pooled across three independent networks, 66.7% versus 40.0% (Fisher exact $p = 0.017$, §3.6, Figure 3), the paper’s one statistically significant positive result, and its clearest evidence that the transport toolkit’s distinctive machinery earns its keep specifically where naive cannot compete.

3.10. What remains open

Naive dominance is depth-dependent, not universal, and the distinction matters for how every other result in this section should be read. It is robust to lattice size at a fixed, shallow threshold depth (§3.7: $L = 8\text{--}32$, no sign of the margin closing), but not to depth itself at fixed size: pushed deep enough that naive fails outright (§3.6), the transport toolkit’s advantage becomes real, and for the self-consistency coordinate, significant. The two scans probe different axes of the same underlying claim and give different answers: treating “naive wins at $L = 16$ ” as a fixed, depth-independent property of the problem would have missed the paper’s one significant positive result entirely.

The learned coordinate trained on milestone committor labels is the clearest case of an apparent win reversing under scrutiny, and remains a genuine loss even after the most thorough attempt in this paper to rescue it. A first training run beat the hand-picked energy coordinate by +14%; seventeen retraining runs of the identical configuration gave outcomes from -52% to $+112\%$ (Table 3), and pooling them left a positive but non-significant point estimate (§3.4). Rather than add more single-net seeds, a deep-ensemble follow-up attacked the diagnosed noise source directly and settled the question in the negative, exposing a cost-accounting gap along the way that turned a marginal-looking loss into this paper’s only confidence interval that excludes parity outright (§3.4.1). The occasional large positive outcomes in the original seed sweep are best read as high-variance artifacts of initialization, not evidence the method sometimes wins.

FW-CADIS’s flattening claim is not established, and the negative result is now well characterized rather than merely observed. Its cost advantage over uniform allocation is real and reproducible ($\sim 8\times$ cheaper for a statistically indistinguishable error profile), but the flattening property it is named for did not appear at any α , replica count, or coordinate tried, and a ferromagnet control (§3.8) rules out spin-glass ruggedness as the explanation: if anything the failure is *worse* on the easier landscape. What remains open is which of two explanations is responsible, the specific bin/threshold parameterisation or a more general limit of inverse-response weighting at this scale; distinguishing them needs a parameter sweep independent of α , not another problem instance or more replicas.

The self-consistency objective’s FOM advantage over this project’s own surrogate is not established at shallow depth (§3.5), and did not visibly grow with depth within the range where naive still competes, though its reliability advantage (fewest zero-replicas of any weight-window method, at every shallow depth tested) held throughout, foreshadowing the significant reliability result the same objective produced once tested where naive genuinely fails (§3.6).

Push-to-scale questions (three dimensions, larger L , deeper than beyond=4) are entirely untouched, and are the natural next test of whether the deep-rare advantage found here continues to grow or plateaus.

3.11. Relation to prior art

Kim & Cai (arXiv:2602.12294, 2026) learn an importance function $I(s)$, modify transitions as $K'(i \rightarrow j) = K(i \rightarrow j)I(j)/I(i)$, and use a branching random walk: in the language of §2.2, their I is our h , their K' is the tilt, and their branching is splitting: they independently reconstruct the zero-variance/weight-window recipe for discrete rare transitions, without the transport framing, and without comparing their learned coordinate against a hand-picked reaction coordinate the way this paper does. We directly implemented and tested their self-consistency training objective (§3.5); what remains open after doing so is not whether it works in principle (their paper demonstrates convergence to the exact transition rate on two test potentials) but whether it offers a measurable practical advantage over simpler alternatives on a problem of this kind at this scale, which our data does not yet support. Doob-transform VAN (Phys. Rev. E 111, 034120, 2025) approximates the leading eigenvector of the tilted generator for *dynamical* large

deviations, the dynamical analogue of $\psi^\dagger$; a head-to-head against a reverse-KL VAN baseline, which this comparison would require, remains untried in this project. No successor work citing either paper closes the specific gap this paper's methods and results address, as of the most recent check.

4. Conclusions

The transport–statistical-mechanics correspondence this project set out to test is real where it can be proved (the adjoint=committor identity), but none of the more experimental claims built on top of it held up as cleanly as an initial pass through the results suggested. The clearest lesson is not any single number: “0/10 zero-replicas” and a tidy point estimate are not, by themselves, evidence of a real effect. FW-CADIS’s flattening claim and the Kim & Cai comparison both needed a bootstrap confidence interval to reveal an apparent win as noise; the milestoning result needed something bootstrapping alone could not supply, independently-trained networks, to reveal the same thing (§3.4, §3.4.1). Reaching these conclusions required diagnosing and correcting real implementation issues along the way: a resampling interval that silently degenerated a weight-window method to naive sampling, a mis-calibrated bin range, an under-determined training objective, and an unseeded network initialization that let a single run be mistaken for a stable result, each caught by an independent cross-check rather than a number simply looking wrong.

Two results stand apart from that pattern, and together they sharpen the paper’s central claim. The milestoning ensemble turned an ambiguous point estimate into this paper’s only confidence interval that excludes parity outright, a decisive negative, not a wash (§3.4.1). The self-consistency objective, meanwhile, whose shallow-depth advantage had itself dissolved under scrutiny and whose training cost is a real practical tax, roughly 170× that of the surrogate objective (§3.5), produced this paper’s one statistically significant positive result once tested at a depth naive genuinely cannot reach (§3.6): the one apparent win in this project that grew *stronger*, not weaker, under the same scrutiny that dissolved every other one. Read together, the transport toolkit’s value is not diffuse across every regime this correspondence touches. It is concentrated specifically in the deep-rare regime radiation transport was built for, and requires a training objective shaped like the actual committor condition rather than a shallow, energy-adjacent proxy of it. We think this is the right way to report a project whose most interesting finding, more than once, was that an apparent win did not survive being checked properly, and whose one win that did survive only appeared once the problem was made hard enough to need it.

Acknowledgements

The author gratefully acknowledges the support provided by Fahad Bin Sultan University, Tabuk, Saudi Arabia, for providing a supportive academic environment and access to the computational facilities necessary to conduct this research. The institutional support and research infrastructure were essential for implementing the Monte Carlo experiments and performing the systematic statistical analysis reported in this work.

CRedit authorship contribution statement

M.Y. Alzaatreh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

References

- [1] H. Kahn and T. E. Harris. Estimation of particle transmission by random sampling. *National Bureau of Standards Applied Mathematics Series*, 12:27–30, 1951.
- [2] H. Kahn and A. W. Marshall. Methods of reducing sample size in Monte Carlo computations. *Journal of the Operations Research Society of America*, 1(5):263–278, 1953.
- [3] R. J. Allen, C. Valeriani, and P. R. ten Wolde. Forward flux sampling for rare event simulations. *Journal of Physics: Condensed Matter*, 21(46):463102, 2009.
- [4] H. Touchette. The large deviation approach to statistical mechanics. *Physics Reports*, 478(1–3):1–69, 2009.
- [5] W. E and E. Vanden-Eijnden. Towards a theory of transition paths. *Journal of Statistical Physics*, 123(3):503–523, 2006.
- [6] P. Metzner, C. Schütte, and E. Vanden-Eijnden. Transition path theory for Markov jump processes. *Multiscale Modeling & Simulation*, 7(3):1192–1219, 2009.
- [7] W. E and E. Vanden-Eijnden. Transition-path theory and path-finding algorithms for the study of rare events. *Annual Review of Physical Chemistry*, 61:391–420, 2010.

- [8] G. A. Huber and S. Kim. Weighted-ensemble Brownian dynamics simulations for protein association reactions. *Biophysical Journal*, 70(1): 97–110, 1996.
- [9] B. W. Zhang, D. Jasnow, and D. M. Zuckerman. The ‘weighted ensemble’ path sampling method is statistically exact for a broad class of stochastic processes and binning procedures. *The Journal of Chemical Physics*, 132:054107, 2010.
- [10] F. Cérou and A. Guyader. Adaptive multilevel splitting for rare event analysis. *Stochastic Analysis and Applications*, 25(2):417–443, 2007.
- [11] P. G. Bolhuis, D. Chandler, C. Dellago, and P. L. Geissler. Transition path sampling: Throwing ropes over rough mountain passes, in the dark. *Annual Review of Physical Chemistry*, 53:291–318, 2002.
- [12] D. M. Zuckerman and L. T. Chong. Weighted ensemble simulation: Review of methodology, applications, and software. *Annual Review of Biophysics*, 46:43–57, 2017.
- [13] J. Lewins. *Importance: The Adjoint Function*. Pergamon Press, Oxford, 1965.
- [14] G. I. Bell and S. Glasstone. *Nuclear Reactor Theory*. Van Nostrand Reinhold, New York, 1970.
- [15] T. E. Booth and J. S. Hendricks. Importance estimation in forward Monte Carlo calculations. *Nuclear Technology/Fusion*, 5, 1984.
- [16] J. C. Wagner and A. Haghighat. Automated variance reduction of Monte Carlo shielding calculations using the discrete ordinates adjoint function. *Nuclear Science and Engineering*, 128(2):186–208, 1998.
- [17] A. Haghighat and J. C. Wagner. Monte Carlo variance reduction with deterministic importance functions. *Progress in Nuclear Energy*, 42(1): 25–53, 2003.
- [18] J. C. Wagner, D. E. Peplow, and S. W. Mosher. FW-CADIS method for global and regional variance reduction of Monte Carlo radiation transport calculations. *Nuclear Science and Engineering*, 176(1):37–57, 2014.
- [19] M. A. Cooper and E. W. Larsen. Automated weight windows for global Monte Carlo particle transport calculations. *Nuclear Science and Engineering*, 137(1):1–13, 2001.
- [20] J. Spanier and E. M. Gelbard. *Monte Carlo Principles and Neutron Transport Problems*. Addison-Wesley, Reading, MA, 1969.
- [21] J. L. Doob. *Classical Potential Theory and Its Probabilistic Counterpart*. Springer-Verlag, New York, 1984.
- [22] R. Chetrite and H. Touchette. Nonequilibrium Markov processes conditioned on large deviations. *Annales Henri Poincaré*, 16(9):2005–2057, 2015.
- [23] C. Hartmann, O. Kebiri, L. Neureither, and L. Richter. Variational approach to rare event simulation using least-squares regression. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 29(6):063107, 2019.
- [24] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [25] Q. Li, B. Lin, and W. Ren. Computing committor functions for the study of rare events using deep learning. *The Journal of Chemical Physics*, 151(5):054112, 2019.
- [26] Y. Khoo, J. Lu, and L. Ying. Solving for high-dimensional committor functions using artificial neural networks. *Research in the Mathematical Sciences*, 6(1), 2019.
- [27] G. M. Rotskoff, A. R. Mitchell, and E. Vanden-Eijnden. Active importance sampling for variational objectives dominated by rare events: Consequences for optimization and generalization. *Proceedings of Machine Learning Research*, 145:757–780, 2022.
- [28] E. Trizio, P. Kang, and M. Parrinello. Everything everywhere all at once: a probability-based enhanced sampling approach to rare events. *Nature Computational Science*, 2025. See also E. Trizio, G. Rossi and M. Parrinello, “Ceci n’est pas un committor, yet it samples like one: efficient sampling via approximated committor functions”, arXiv:2602.23236, 2026.
- [29] D. Wu, L. Wang, and P. Zhang. Solving statistical mechanics using variational autoregressive networks. *Physical Review Letters*, 122:080602, 2019.
- [30] F. Noé, S. Olsson, J. Köhler, and H. Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457), 2019.
- [31] K. A. Nicoli et al. Asymptotically unbiased estimation of physical observables with neural samplers. *Physical Review E*, 101:023304, 2020.
- [32] M. Gabrié, G. M. Rotskoff, and E. Vanden-Eijnden. Adaptive Monte Carlo augmented with normalizing flows. *Proceedings of the National Academy of Sciences*, 119(10):e2109420119, 2022.
- [33] E. Ising. Beitrag zur Theorie des Ferromagnetismus. *Zeitschrift für Physik*, 31(1):253–258, 1925.
- [34] L. Onsager. Crystal statistics. I. A two-dimensional model with an order-disorder transition. *Physical Review*, 65(3–4):117–149, 1944.
- [35] S. F. Edwards and P. W. Anderson. Theory of spin glasses. *Journal of Physics F: Metal Physics*, 5(5):965–974, 1975.
- [36] K. Binder and A. P. Young. Spin glasses: Experimental facts, theoretical concepts, and open questions. *Reviews of Modern Physics*, 58(4): 801–976, 1986.
- [37] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953.